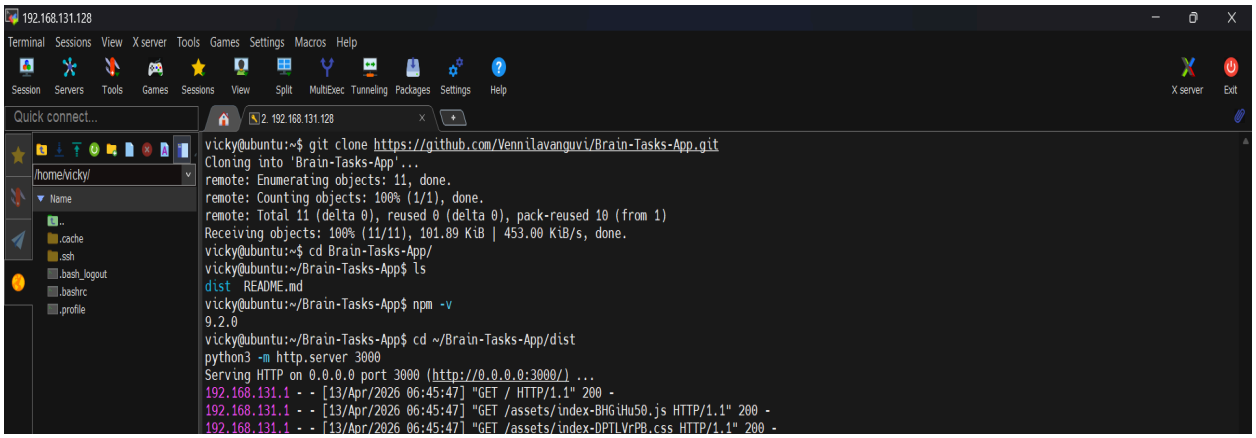
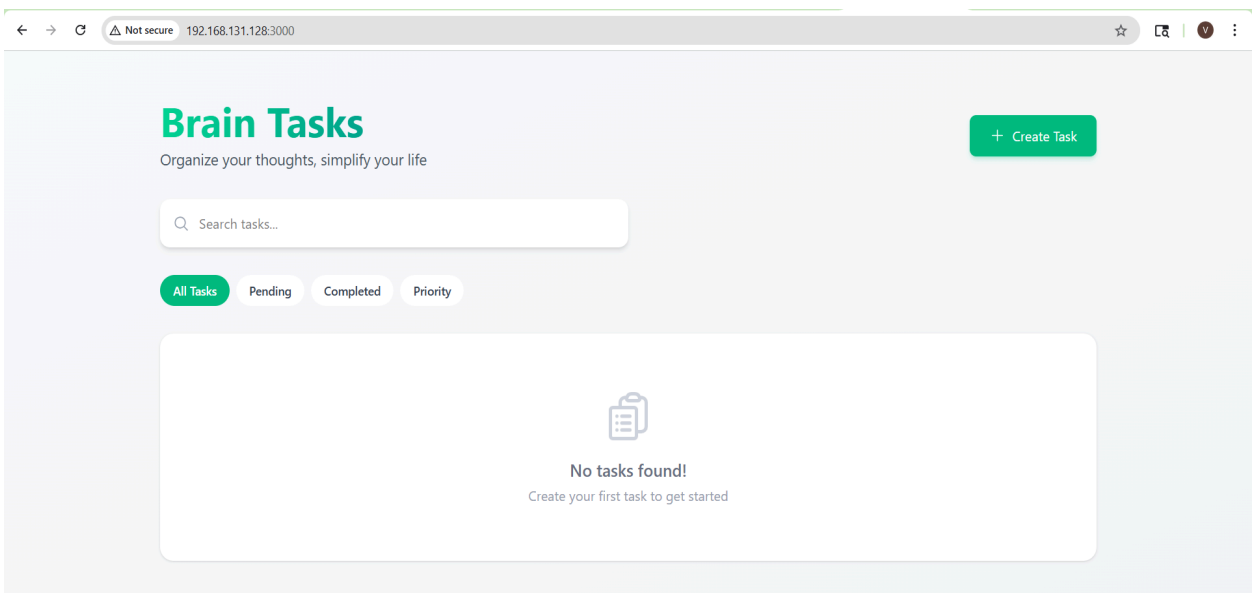
Brain Task Application

Git Repository :

Repository : https://github.com/wikkics/Brain-Tasks-App
Region : us-east-1 (N. Virginia)

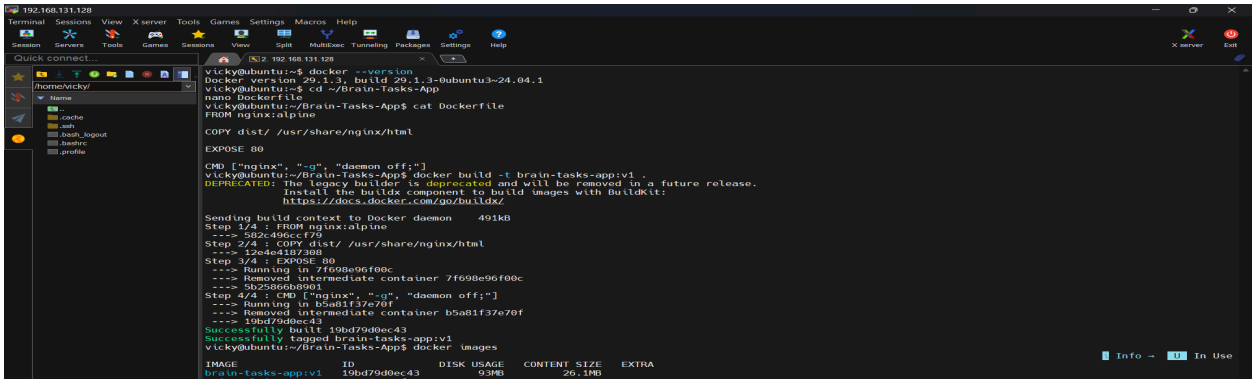
Cloned the Brain Tasks App repository and verified the application files locally before containerizing.

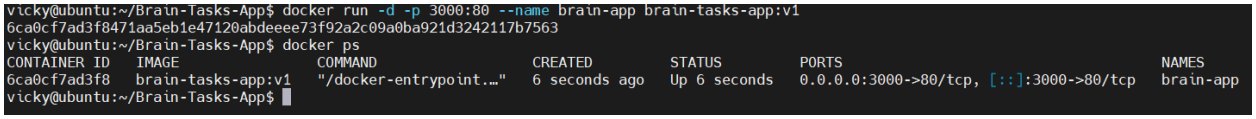


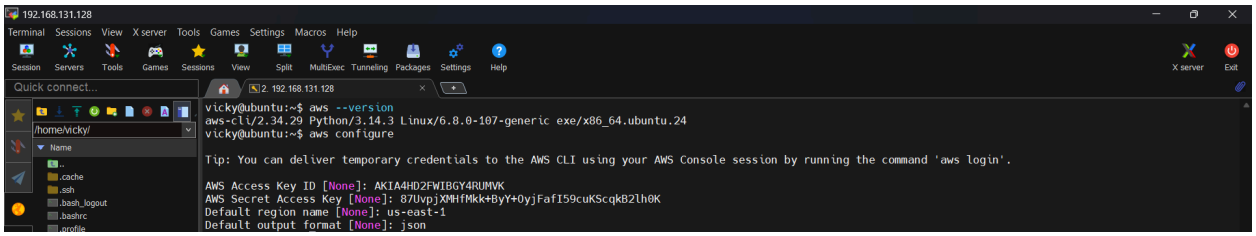


Dockerize the Application :

Created a Dockerfile to package the application with Nginx web server into a portable Docker image. Used AWS ECR Public for the Nginx base image to avoid DockerHub rate limits in CI/CD

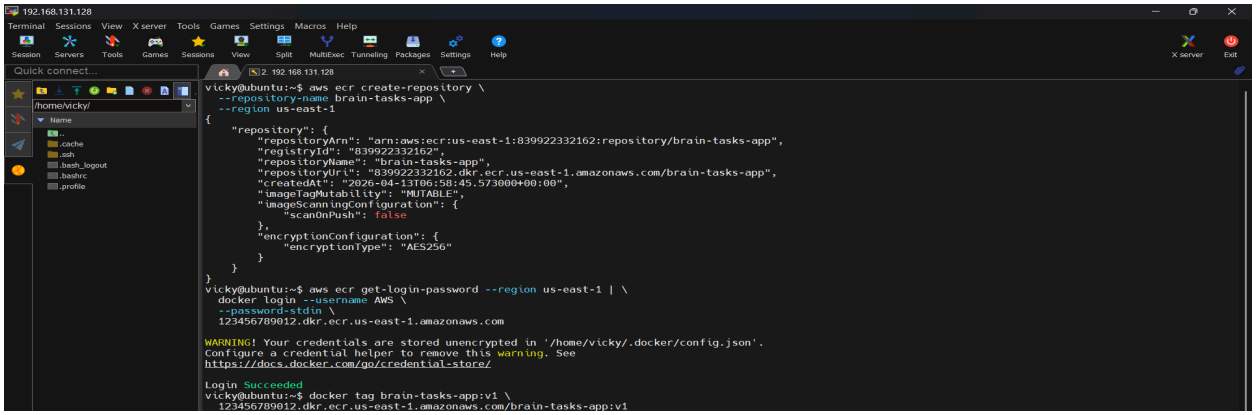






Container Registry (AWS ECR):

Created an AWS ECR repository and pushed the Docker image to the cloud so Kubernetes can pull it during deployment. AWS ECR is a fully managed private Docker registry — faster, more secure (IAM-controlled), and free of DockerHub rate limits



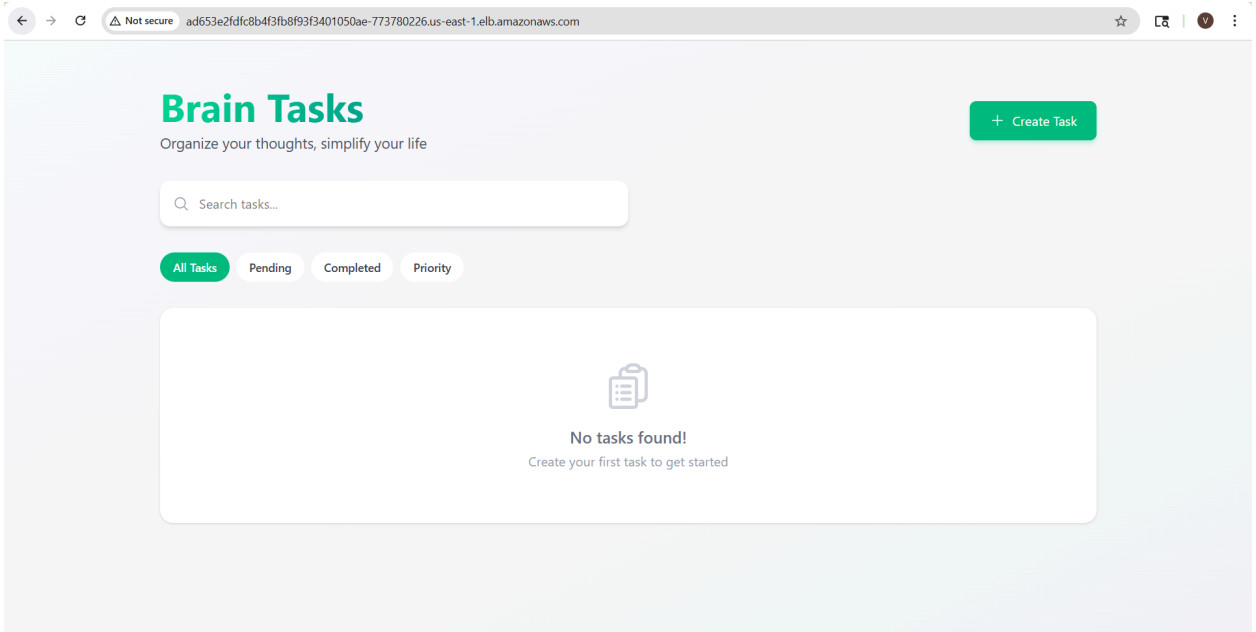
```
The push refers to repository [839922332162.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app]
00d3d76a730b: Pushed
34dfdd2ef1f9: Pushed
c8a2fa3a88d2: Pushed
589002ba0eae: Pushed
f03becc8ac15: Pushed
15e759724ff6: Pushed
ff9f59a6a62e: Pushed
a71873b303e8: Pushed
1165b869c51a: Pushed
v1: digest: sha256:19bd79d0ec43bad6b8cbdbe684889af3ae5d8068484bc8b202394b448d2a99ab size: 2264
vicky@ubuntu:~$ aws ecr list-images \
  --repository-name brain-tasks-app \
  --region us-east-1
{
    "imageIds": [
        {
            "imageDigest": "sha256:19bd79d0ec43bad6b8cbdbe684889af3ae5d8068484bc8b202394b448d2a99ab",
            "imageTag": "v1"
        }
    ]
```

Kubernetes on AWS EKS :

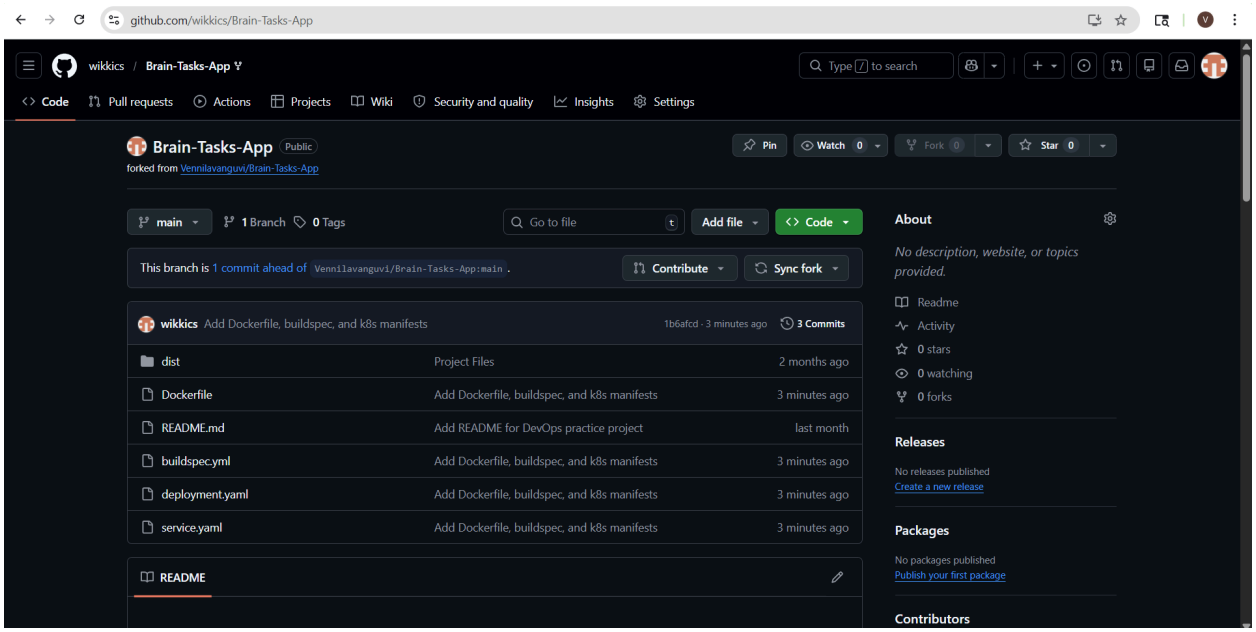
Created a managed Kubernetes cluster on AWS EKS with a t3.small worker node, wrote Kubernetes manifests, and deployed the app with a public LoadBalancer URL.

```
kubectl get nodes
Updated context arn:aws:eks:us-east-1:839922332162:cluster/brain-tasks-cluster in /home/vicky/.kube/config
NAME                                       STATUS   ROLES    AGE     VERSION
ip-192-168-14-201.ec2.internal   Ready    <none>   7m17s   v1.34.4-eks-f69f56f
vicky@ubuntu:~/Brain-Tasks-App$ ls
deployment.yaml  dist  Dockerfile  README.md  service.yaml
vicky@ubuntu:~/Brain-Tasks-App$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: brain-tasks-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: brain-tasks
  template:
    metadata:
      labels:
        app: brain-tasks
    spec:
      containers:
      - name: brain-tasks
        image: 839922332162.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:v1
        ports:
        - containerPort: 80
vicky@ubuntu:~/Brain-Tasks-App$ cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: brain-tasks-service
spec:
  selector:
    app: brain-tasks
  type: LoadBalancer
  ports:
  - protocol: TCP
    port: 80
    targetPort: 80
```

```
vicky@ubuntu:~/Brain-Tasks-App$ echo "=== Nodes ===" && \
kubectl get nodes && \
echo "=== Pods ===" && \
kubectl get pods && \
echo "=== Deployment ===" && \
kubectl get deployment && \
echo "=== Service ===" && \
kubectl get service brain-tasks-service
=== Nodes ===
NAME                                       STATUS   ROLES    AGE   VERSION
ip-192-168-14-201.ec2.internal   Ready    <none>   11m   v1.34.4-eks-f69f56f
=== Pods ===
NAME                                     READY   STATUS    RESTARTS   AGE
brain-tasks-deployment-f65d9bd97-chqkl   1/1     Running   0          2m35s
=== Deployment ===
NAME                   READY   UP-TO-DATE   AVAILABLE   AGE
brain-tasks-deployment   1/1     1            1           2m39s
=== Service ===
NAME                 TYPE           CLUSTER-IP       EXTERNAL-IP                                                              PORT(S)        AGE
brain-tasks-service   LoadBalancer   10.100.209.141   ad653e2fdfc8b4f3fb8f93f3401050ae-773780226.us-east-1.elb.amazonaws.com   80:32033/TCP   2m34s
vicky@ubuntu:~/Brain-Tasks-App$
```

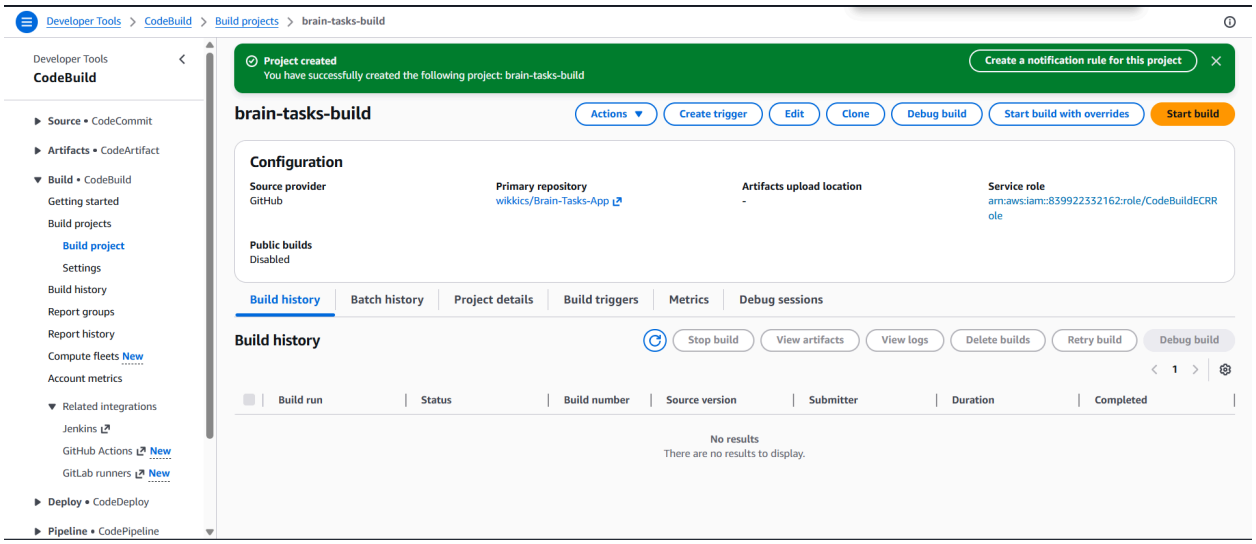


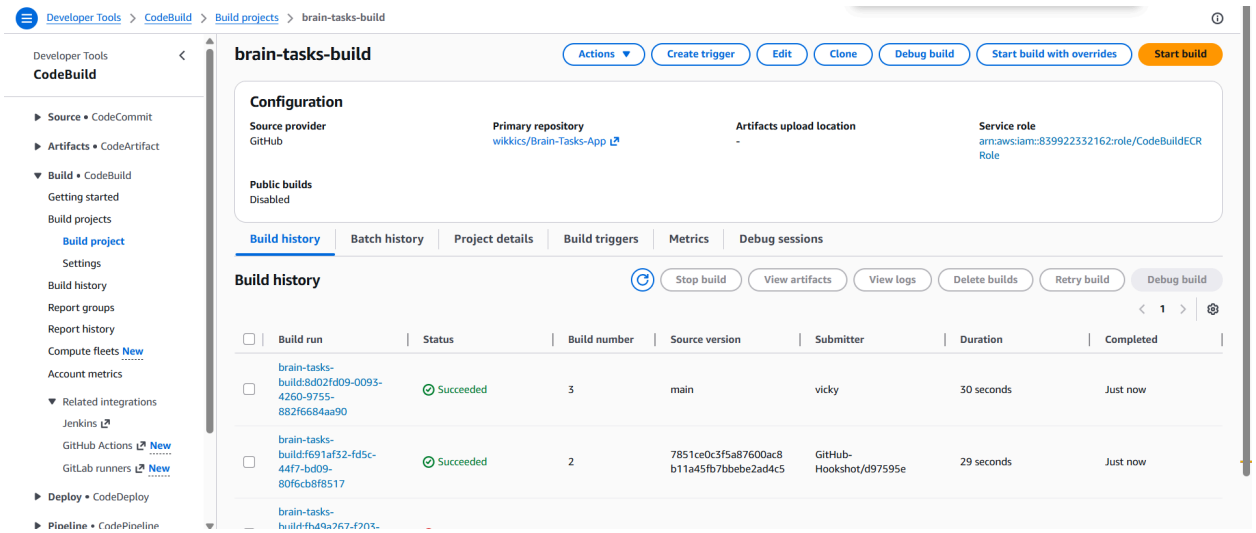

```
vicky@ubuntu:~/Brain-Tasks-App$ cd ~/Brain-Tasks-App

# Change remote to your forked repo
git remote set-url origin https://github.com/wikkics/Brain-Tasks-App.git

# Verify
git remote -v
origin  https://github.com/wikkics/Brain-Tasks-App.git (fetch)
origin  https://github.com/wikkics/Brain-Tasks-App.git (push)
vicky@ubuntu:~/Brain-Tasks-App$ git push origin main
Username for 'https://github.com': wikkics
Password for 'https://wikkics@github.com':
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 1.14 KiB | 290.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/wikkics/Brain-Tasks-App.git
   d611aeb..1b6afcd  main -> main
vicky@ubuntu:~/Brain-Tasks-App$
```

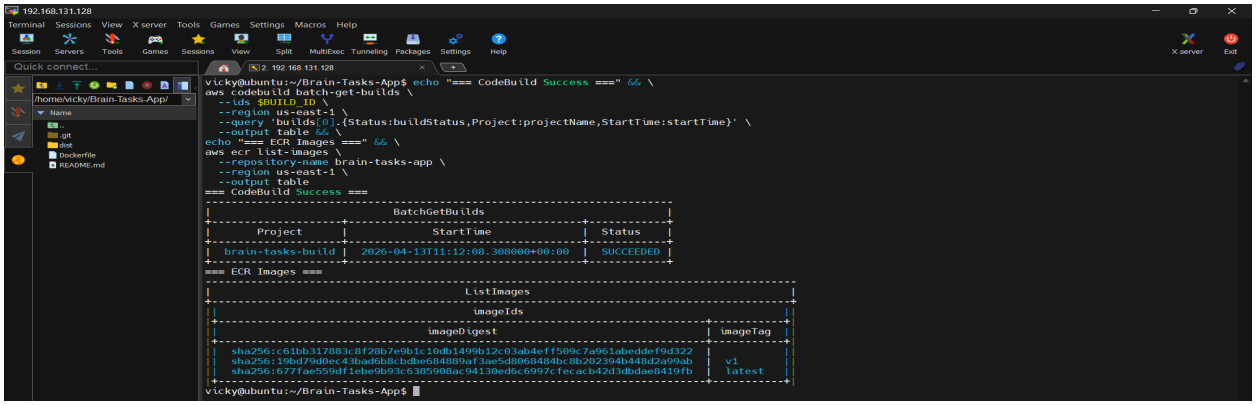


AWS CodeBuild (CI) :

Created a CodeBuild project that automatically builds the Docker image, pushes to ECR, and deploys to EKS. The buildspec.yml defines all build instructions.

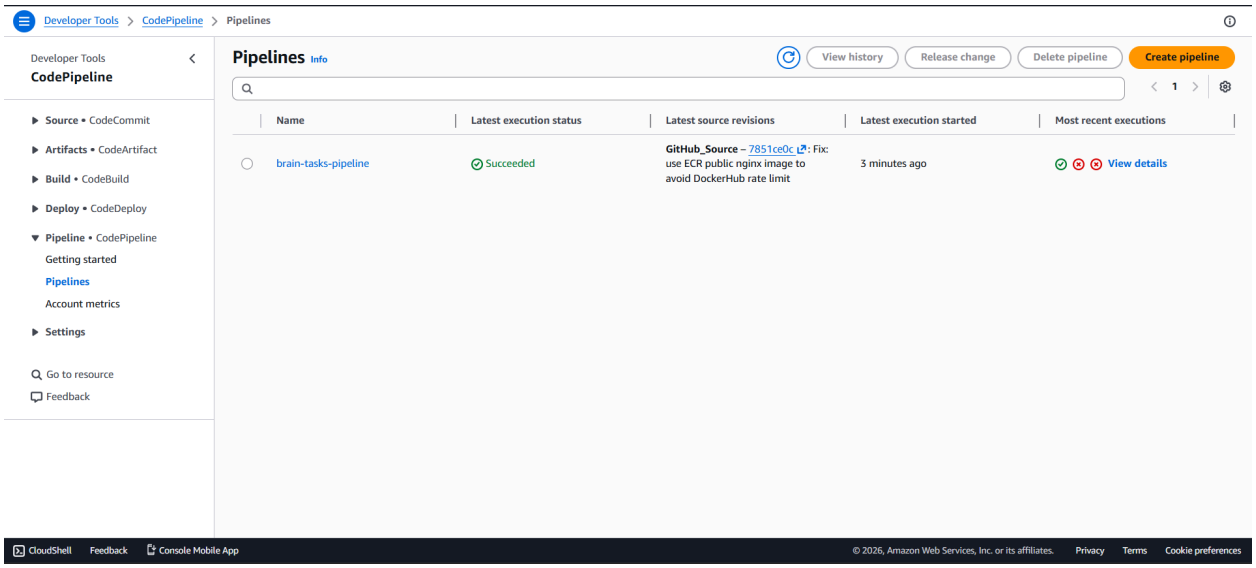


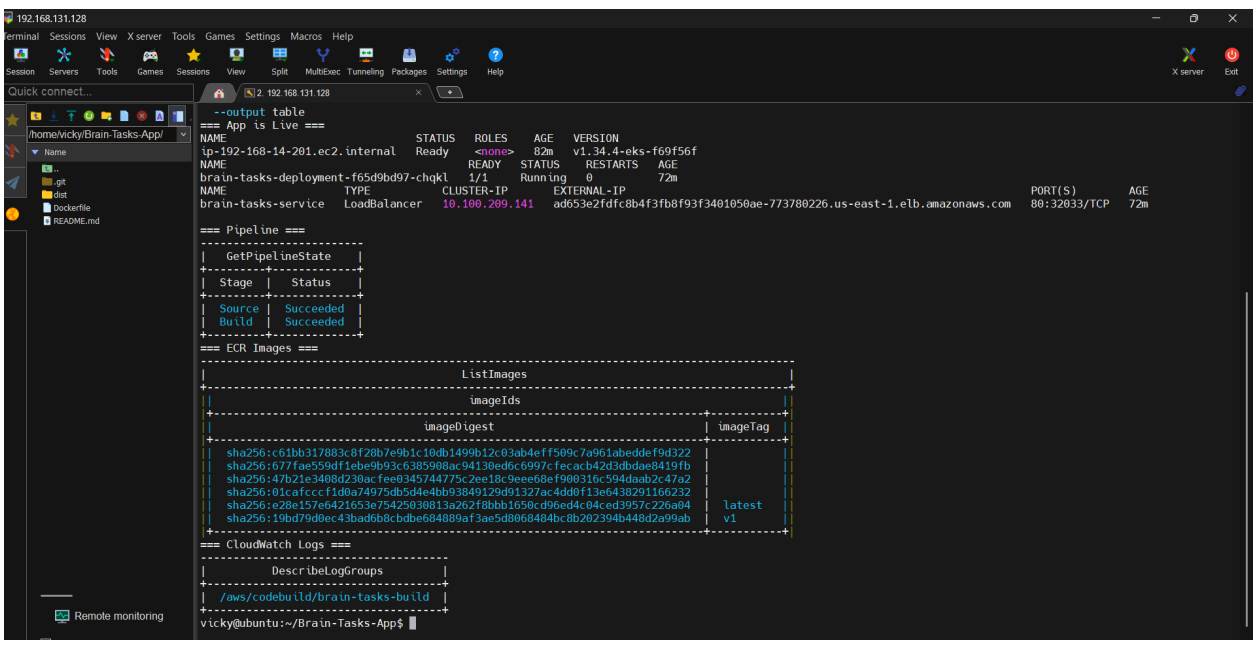
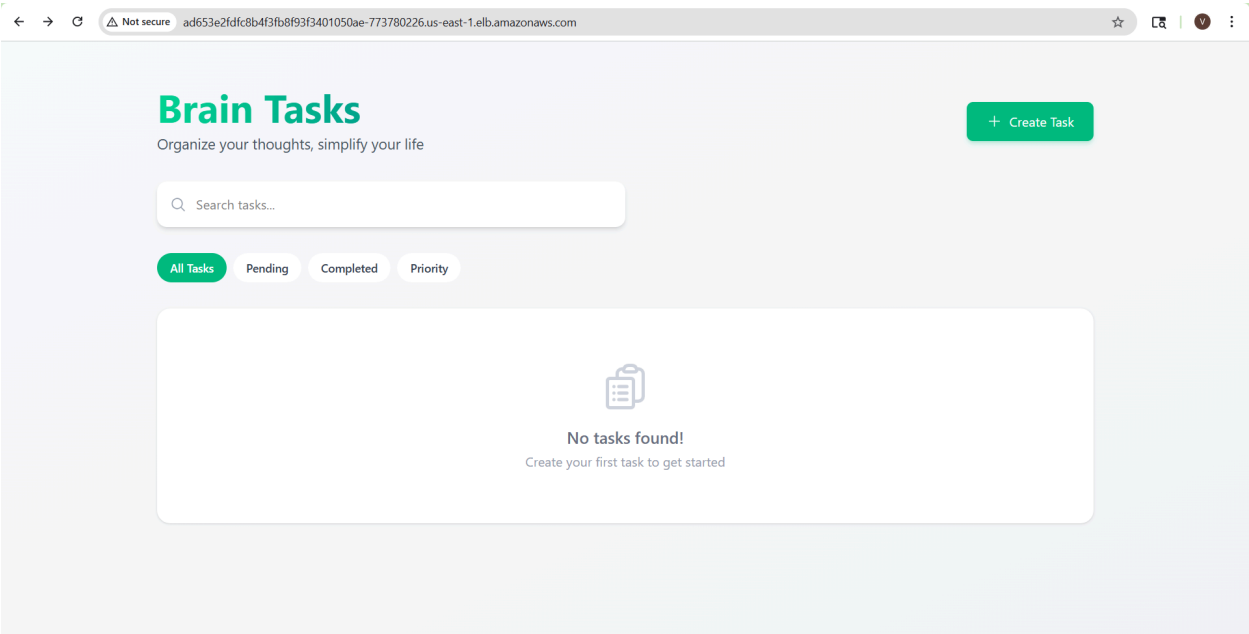




AWS CodePipeline (CD) :

Created a two-stage pipeline connecting GitHub to CodeBuild. Every push to the main branch automatically triggers a full build and deploymen





Field	Details
Project Name	Brain Tasks App
Repository	https://github.com/wikkics/Brain-Tasks-App
Application Type	React Web Application
Deployment Platform	Kubernetes (AWS EKS / EC2-based cluster assumed)
Containerization	Docker
CI/CD Tool	AWS CodeBuild
Web Server	Nginx
Live URL	http://ad653e2fdfc8b4f3fb8f93f3401050ae-773780226.us-east-1.elb.amazonaws.com